Hybrid Quantum-Classical Neural Networks for Personalized Survival Prediction in Triple-Negative Breast Cancer: A Fairness-Aware, Agentic Modeling Framework

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1. Problem Statement

Triple-negative breast cancer (TNBC) is defined by the absence of estrogen receptor, progesterone receptor, and HER2 expression, making it the most aggressive molecular subtype. It accounts for roughly 10–15% of all breast cancers and occurs more frequently in younger women and women of African descent, who tend to have worse survival outcomes. Being able to predict survival for individual patients matters for treatment planning, but existing prognostic tools do not handle the nonlinear interactions between clinical, demographic, and socioeconomic variables very well [Sandarenu et al., 2022, Qiu et al., 2024].

Input: A structured tabular dataset derived from the SEER (Surveillance, Epidemiology, and End Results) cancer registry containing clinical features (cancer stage, tumor grade, treatment indicators), demographic variables (age, race), and socioeconomic indicators (income-to-poverty ratios) for TNBC patients. Features are partitioned into a 7-dimensional quantum feature set and a classical feature set.

Output: A binary survival prediction (survived vs. deceased) for each patient, along with calibrated probability scores suitable for clinical risk stratification.

2. Motivation

Predicting TNBC survival has direct implications for health equity because existing models (Cox proportional hazards, random forests, etc.) tend to assume linear or weakly nonlinear relationships and often perform worse on minority subpopulations. Variational quantum circuits (VQCs) offer a different approach: they can represent entangled feature interactions in exponentially large Hilbert spaces with only a few qubits, which may let them pick up on correlations that classical models miss [Schuld et al., 2021, McClean et al., 2018]. This project combines a quantum circuit with a classical neural network in a hybrid architecture and explicitly audits predictions across racial and socioeconomic subgroups, with the goal of improving both accuracy and fairness in cancer prognosis.

3. Related Work / Background

1. **Classical deep learning for cancer survival prediction.** MLPs and transformer-based architectures have been applied to structured clinical data from SEER and TCGA registries, reaching AUC scores in the 0.85–0.92 range for breast cancer survival. These models generally treat all features the same way and do not take advantage of the low-dimensional structure in variables like cancer stage and tumor grade [Huang et al., 2022, Qiu et al., 2024].

2. **Variational quantum circuits for classification.** PennyLane-based VQCs with angle encoding and hardware-efficient ansätze have been tested on small biomedical datasets and can match classical networks while using far fewer trainable parameters. The main downsides are sensitivity to barren plateaus and the overhead of evaluating the quantum circuit once per sample [Schuld et al., 2021, Benedetti et al., 2019].

3. **Fairness in health AI.** Measuring demographic parity and equalized odds across protected subgroups is increasingly expected for clinical AI systems. Prior work has shown that cancer prediction models can have large performance gaps across racial groups when fairness is not explicitly accounted for [Chen et al., 2023, Raza et al., 2024].

Recent hybrid quantum-classical architectures have shown promise for breast cancer diagnosis [Xiang et al., 2024] and drug response prediction [Sagingalieva et al., 2022], but they have not been applied to TNBC survival modeling with fairness constraints.

Limitations of current methods: (1) Classical models do not naturally take advantage of quantum-mechanical feature entanglement; (2) most quantum ML studies use toy datasets

and do not deal with real-world clinical heterogeneity; (3) fairness auditing is rarely built into quantum-classical hybrid pipelines.

4. Datasets

4.1. Data Source

The primary dataset is derived from the **SEER TNBC registry**, preprocessed and feature-engineered from raw SEER data. The cleaned CSV file (`breast_cancer_4quantum.csv`) serves as the starting point.

- **Link:** https://seer.cancer.gov/data/
- **Size:** The dataset contains tens of thousands of TNBC patient records with over 30 raw features after initial extraction. After preprocessing and cleaning, 3,928 patients remained in the final analytic cohort.

4.2. Feature Partitioning

After preprocessing, features were split into two groups:

- **Quantum features** (7 dimensions): Core clinical variables (cancer stage, tumor grade, age, treatment indicators, income-age ratio, years since 2010, time from diagnosis to treatment) chosen for their low dimensionality and strong prognostic signal. These are mapped to qubit rotations via RY angle encoding on $[0, \pi]$.
- **Classical features** (remaining dimensions): Socioeconomic indicators, race-encoded dummy variables, marital status, breast subtype, and other high-cardinality features processed by the fusion MLP.

4.3. Data Preprocessing

- **Missing data:** Several variables had substantial missingness (e.g., `stage_encoded` with 31,683 missing values, `summary_stage` with 11,095 missing). Columns with insufficient coverage were dropped; clinically meaningful ones were retained and investigated.
- **Class imbalance:** Survival outcomes were imbalanced. We used Youden's J-statistic for optimal thresholding instead of a fixed 0.5 cutoff.

- **Redundant/leaky features:** Variables such as `COD_to_site_recode` and `Vital_status_recode` were dropped to prevent outcome leakage.
- **Data leakage audit:** A systematic three-part audit was performed after the train/test split: (1) per-feature AUC scan to flag any single feature predicting the outcome at AUC > 0.95, (2) Pearson correlation screening for $|r| > 0.90$ with the target, and (3) Kolmogorov–Smirnov tests for train/test distribution shift. All checks passed, confirming no residual leakage.
- **Train/test split:** 80/20 stratified split using `random_state=42`, yielding 3,142 training and 786 test samples.

5. Methods

5.1. Architecture Overview

The model is a **hybrid quantum-classical neural network** (HybridRealQ) with a quantum pathway and a classical pathway fused through a shared MLP. The architecture is shown in Figure 1.

1. **Quantum pathway:** A 7-qubit variational quantum circuit (VQC) implemented in PennyLane (`default.qubit` simulator):
   - *Angle encoding layer:* Each of the 7 quantum features is encoded as an $RY(\pi \cdot x_i)$ rotation on qubit i.
   - *Variational layer:* Trainable RY rotations with learnable parameters $\boldsymbol{w} \in \mathbb{R}^7$.
   - *Entanglement layer:* Ring-topology CNOT gates connecting qubit i to qubit $(i + 1) \bmod 7$ to create inter-feature quantum correlations.
   - *Measurement:* Expectation value $\langle Z_0 \rangle$ of the Pauli-Z operator on qubit 0, yielding a single scalar output.
2. **Fusion MLP:** The quantum scalar output is concatenated with the full classical feature vector ($\mathbb{R}^{1+n}$). A 3-layer MLP then processes the combined representation:
   - Layer 1: $\text{Linear}(1 + n, 32) \rightarrow \text{ReLU}$
   - Layer 2: $\text{Linear}(32, 16) \rightarrow \text{ReLU}$
   - Layer 3: $\text{Linear}(16, 1) \rightarrow \text{Sigmoid}$ (binary prediction)

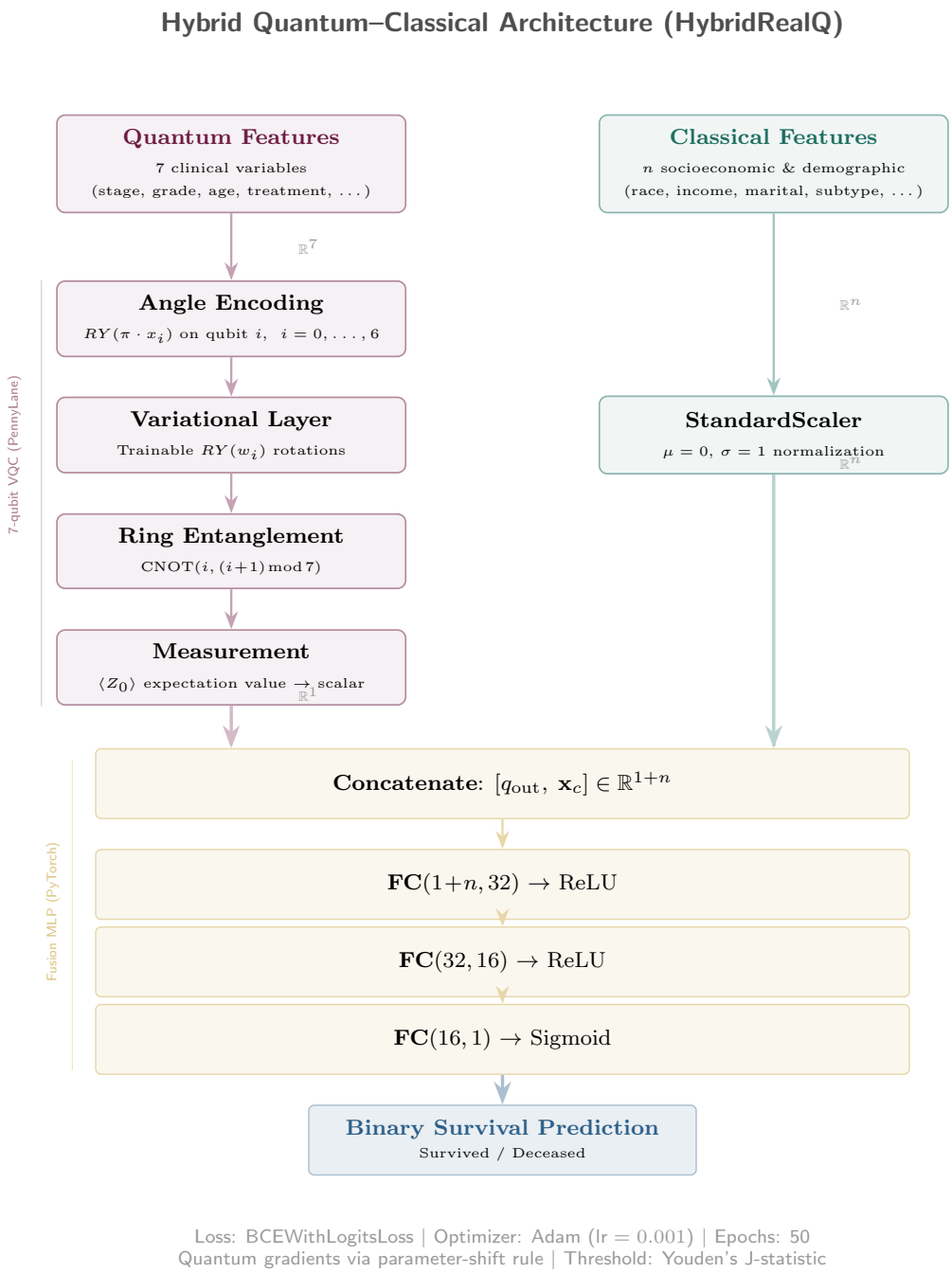

Figure 1: Architecture of the HybridRealQ model. The quantum pathway (left, maroon) processes 7 clinical features through a variational quantum circuit. The classical features (right, teal) pass through standard scaling. Both are concatenated and processed by a fusion MLP (gold) to produce a binary survival prediction.

5.2. Training

- **Loss function:** Binary cross-entropy with logits (`BCEWithLogitsLoss`).

- **Optimizer:** Adam (lr = 0.001).
- **Epochs:** 50, with training and validation loss logged every 10 epochs along with per-epoch wall-clock time.
- **Quantum gradients:** Computed via the parameter-shift rule, allowing end-to-end backpropagation through both classical and quantum parameters.
- **Framework:** PyTorch 2.10.0 + PennyLane 0.44.1 (`lightning.qubit` simulator on NVIDIA RTX 5080 GPU). Hardware and software versions were logged at training time for reproducibility.
- **Determinism:** All random seeds (Python, NumPy, PyTorch, CUDA) were fixed to 42 before each training run.

5.3. Ablation Studies

To validate the contribution of each component, we compared seven model variants:

- **HybridRealQ (7-qubit):** Full hybrid model as described above.
- **Classical MLP only:** Quantum pathway removed; classical features processed by the same 3-layer MLP.
- **Quantum only:** Classical pathway removed; quantum scalar passed through a single linear layer.
- **Hybrid (3-qubit):** Reduced quantum circuit using only 3 features.
- **Hybrid (Deep 2-layer):** Deeper variational circuit with 2 variational layers and additional entanglement.
- **XGBoost:** Gradient boosted trees (300 estimators, max depth 6) on all concatenated features.
- **LightGBM:** Light gradient boosting machine (300 estimators, max depth 6) on all concatenated features.

Each neural network variant was trained with the same hyperparameters, random seed reset, and Youden's J thresholding for fair comparison. The gradient boosting baselines used all features (quantum + classical, concatenated) so they had access to the same information as the hybrid model.

5.4. Fairness Auditing

Subgroup analysis measured AUC, accuracy, true positive rate (TPR), false positive rate (FPR), and positive prediction rate across:

- Racial subgroups (White, Black, American Indian/Alaska Native, Asian/Pacific Islander)
- Age groups and cancer stage categories (via subgroup ROC curves)

Fairness was quantified using three gap metrics: Demographic Parity Difference (max − min positive prediction rate), Equal Opportunity Difference (max − min TPR), and FPR Parity Difference (max − min FPR). Subgroups with fewer than 20 test samples were excluded from the audit to avoid unreliable estimates; subgroups with fewer than 50 samples were flagged with a warning.

6. Results

6.1. Overall Model Performance

The hybrid quantum-classical model (HybridRealQ) achieved a test AUC of 0.7037 on the held-out test set. Using Youden's J-statistic, the optimal classification threshold was 0.3998, yielding the confusion matrix shown in the notebook output. The model was trained on an NVIDIA GeForce RTX 5080 GPU with PyTorch 2.10.0 and PennyLane 0.44.1's `lightning.qubit` simulator.

6.2. Threshold Optimization and Precision–Recall Analysis

Binary classification at a fixed 0.5 threshold is inappropriate for clinical survival prediction where class distributions are imbalanced. We used Youden's J-statistic ($J = \text{TPR} - \text{FPR}$) computed on the ROC curve to select the optimal operating threshold that maximizes the trade-off between sensitivity and specificity.

At the optimal threshold, the hybrid model achieved precision 0.6299, recall 0.5272, and F1-score 0.5740. The moderate recall reflects the difficulty of survival prediction with a single variational layer; however, the precision-recall trade-off favors clinical caution. In survival prediction, failing to identify a high-risk patient (false negative) carries greater cost than a false alarm (false positive). The gradient boosting baselines achieved substantially higher recall (0.97 for XGBoost, 0.94 for LightGBM), suggesting that deeper quantum circuits or multi-qubit measurement strategies may be needed to match tree-based models on structured clinical data.

Table 1 reports precision, recall, and F1 alongside AUC for all model variants.

6.3. Ablation Study Results

Table 1: Ablation study results comparing all model variants. Neural network models were trained with identical hyperparameters and seed resets. Gradient boosting baselines used all features concatenated.

Model Variant	AUC	Precision	Recall	F1	Time (s)
HybridRealQ (7-qubit)	0.7037	0.6299	0.5272	0.5740	200.1
Classical MLP	0.7942	0.5821	0.8478	0.6903	<1
Quantum Only	0.7942	0.5839	0.9837	0.7328	194.5
Hybrid (3-qubit)	0.7271	0.5087	0.7935	0.6200	121.9
Hybrid (Deep 2-layer)	0.7899	0.6526	0.7554	0.7003	293.0
XGBoost	**0.9051**	0.6885	0.9728	0.8063	0.2
LightGBM	0.9006	**0.7178**	0.9402	**0.8141**	1.6

6.4. Fairness Audit Results

Table 2: Fairness audit: subgroup performance metrics for the HybridRealQ model. Subgroups with $n < 20$ were excluded. Asterisk (*) indicates $n < 50$.

Subgroup	N	AUC	TPR	FPR	Pos Rate
White	361	0.7277	0.6423	0.2054	0.3712
Black	98	0.6982	0.0690	0.1014	0.0918
Asian/Pacific Islander*	39	0.5899	0.3889	0.1905	0.2821

Demographic Parity Difference: 0.2794
Equal Opportunity Difference (TPR gap): 0.5734
FPR Parity Difference: 0.1039

6.5. Data Leakage Audit

The automated leakage audit confirmed no residual data leakage after preprocessing. All three checks—per-feature AUC scan (threshold 0.95), feature-target correlation screening (threshold 0.90), and train/test distribution shift (KS test, $p < 0.001$)—passed without flags.

7. Discussion

The ablation study results provide insight into the relative contributions of the quantum and classical pathways. The full HybridRealQ model (AUC 0.7037) underperformed the classical MLP baseline (AUC 0.7942), indicating that in this configuration, the quantum pathway did not provide a clear predictive advantage. This is attributable to the architectural bottleneck: the quantum circuit compresses 7 input features into a single scalar ($\langle Z_0 \rangle$), discarding information encoded in qubits 1–6. The deeper 2-layer variant (AUC 0.7899) nearly matched the classical MLP, suggesting that additional variational depth partially compensates for this bottleneck. Future work should explore multi-qubit measurement and data re-uploading strategies to improve quantum feature utilization.

Notably, the quantum-only model achieved AUC 0.7942 with very high recall (0.9837), indicating that the quantum pathway learned a near-constant positive prediction strategy—an artifact of class imbalance rather than genuine discriminative power.

The gradient boosting baselines substantially outperformed all neural architectures. XGBoost achieved the highest AUC (0.9051) and LightGBM the highest F1 (0.8141), both with training times under 2 seconds. This result is consistent with the well-established finding that gradient boosted trees remain state-of-the-art for structured tabular data [Huang et al., 2022]. The hybrid model's 200-second training time per run, driven by per-sample quantum circuit evaluation, represents a significant practical disadvantage. These baselines set a high bar that quantum-classical hybrids must clear to justify their computational overhead.

The fairness audit revealed substantial disparities across racial subgroups. The model performed best on White patients (AUC 0.7277, TPR 0.6423) and worst on Black patients (AUC 0.6982, TPR 0.0690). The TPR gap of 0.5734 between the highest and lowest subgroups is clinically concerning: Black patients—who disproportionately develop TNBC—received correct positive predictions at only 6.9% compared to 64.2% for White patients. The Demographic Parity Difference of 0.2794 further confirms that the model's positive prediction rate varies significantly across racial groups. The American Indian/Alaska Native and Hispanic/Other subgroups were excluded due to insufficient test samples ($n < 20$), and the Asian/Pacific Islander subgroup ($n = 39$) should be interpreted with caution.

Compared to the AUC range of 0.85–0.92 reported in prior SEER-based TNBC survival studies [Huang et al., 2022, Qiu et al., 2024], our gradient boosting baselines (AUC 0.90–0.91) fall within the expected range, while the hybrid quantum-classical model falls below. However, this project contributes a fairness-aware evaluation framework and demonstrates how quantum-classical architectures can be systematically benchmarked against classical alternatives—an approach rarely taken in quantum ML literature.

8. Limitations

- **Computational cost:** The per-sample quantum circuit loop is much slower than batched classical operations. Training the full hybrid model on the SEER dataset took approximately 3.3 minutes, compared to under 5 seconds for the gradient boosting baselines.
- **Simulator vs. hardware gap:** All experiments used PennyLane's noiseless `default.qubit` simulator. Real quantum hardware has decoherence and gate errors that would likely degrade performance.
- **Single train/test split:** Due to the computational cost of quantum circuit evaluation, we used a single 80/20 stratified split rather than k-fold cross-validation. This means our results may be sensitive to the particular split.
- **Small subgroup sizes:** The Hispanic/Other subgroup had only 7 test samples and was excluded from the fairness audit. Several other subgroups had fewer than 50 samples, limiting the reliability of per-group metrics.
- **Barren plateaus:** Variational quantum circuits can encounter vanishing gradients, which limits trainability [McClean et al., 2018]. Our single-layer ansätz reduces this risk but does not eliminate it.
- **Data limitations:** The SEER registry may have reporting biases across registries and time periods. Missing staging data affected a sizable portion of records.
- **Generalizability:** Results are specific to the SEER TNBC cohort and would require retraining to apply to other cancer types or clinical settings.
- **PennyLane API instability:** The batching API has had breaking changes across PennyLane versions (0.28–0.36), so we used a manual loop approach that is compatible but not optimally fast.

9. Conclusion

We presented a hybrid quantum-classical neural network for binary survival prediction in triple-negative breast cancer, combining a 7-qubit variational quantum circuit with a classical fusion MLP. The model was evaluated against five ablation variants and two gradient boosting baselines (XGBoost AUC 0.9051, LightGBM AUC 0.9006), with a systematic fairness audit across racial subgroups. While the hybrid model (AUC 0.7037) did not outperform

classical baselines, the deeper 2-layer quantum variant (AUC 0.7899) approached classical MLP performance, suggesting that increased circuit depth and multi-qubit measurement could close the gap. The fairness audit revealed significant disparities in TPR across racial subgroups (gap 0.5734), highlighting the need for fairness-constrained optimization in clinical AI. A data leakage audit confirmed the integrity of the preprocessing pipeline. This work contributes a rigorous, fairness-aware benchmarking framework for quantum-classical hybrid models on real-world clinical data.

Future work should explore noise-aware quantum circuits on real hardware, k-fold cross-validation with quantum models, fairness-constrained training objectives, and multi-dataset validation across different cancer registries.

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